

EC 603 ADVANCED MACROECONOMICS I Fall 2019

PART I

Instructor: Murat Koyuncu

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PART II

Instructor: Meltem Poyraz

Time and Location: W234 10:00-12:50, NB Z13

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Prerequisites

EC 503 Macroeconomics I and EC 504 Macroeconomics II (master's level core courses in the macroeconomics sequence) are the official prerequisites of this course.

Course requirements

Class participation

You are expected to attend all lectures and participate in discussions.

Assignments

You are required to post three questions for each paper that will be presented that week. A selection will be discussed in class. The questions should be sent to the professor via e-mail by 22:00 of the day before the class at the latest.

Paper Presentation

Each student will be assigned two or three papers from the reading list provided below to present in class.

Project Paper

A major element of this course is the completion of a project paper. This exercise is intended to get students started thinking about a particular topic and conduct an extensive research on it. The project paper should not be a repository of the literature, it must have an original contribution to the literature. This may sound daunting right now, but it is exactly what you need to get started on something.

In the last lecture of the course (on Dec 18th) you will present your project topic and progress to get feedback from us and your classmates before the final submission. Deadline for the paper will be the final date of the course determined by the Registration Office. The paper should be written in a clear form submittable to the Economics Bulletin. Details on this will be discussed in class.

Grading Policy and Grading Scale

The weights for the overall score will be as follows:

Part 1	Percentage of final grade
Participation	10%
Presentations	20%
Part 2	
Participation	10%
Presentations	20%
End of Semester	
Project Paper	40%

The standard catalog grading protocol accepted at Boğaziçi University is as follows:

Grade	Interpretation	Weight	Grade
AA	Excellent	4.0	90%
BA	Good-Excellent	3.5	85-89%
BB	Good	3.0	80-84%
CB	Passing-Good	2.5	75-79%
CC	Passing	2.0	70-74%
F	Failed	0.0	0-59%

Communication

E-mail will be used as the primary means of communication outside the classroom, and we will be sending e-mails whenever necessary to inform you on updates regarding the course. Please make sure that the e-mail address you have at the Registrar's Office is up-to-date so that you will not miss any of the announcements.

Academic Integrity:

The undergraduate program of the Department of Economics is conducted within the framework of the Student Discipline Regulations of the Turkish Council of Higher Education (TCHE, Yükseköğretim Kurulu, YÖK in Turkish), and rules accepted by the Boğaziçi University Committee on Ethical Conduct.

For the student discipline regulations by the TCHE (YÖK), see (in Turkish):

<http://bit.ly/yokdisiplinyonetmeligi>

For the ethics code accepted by Boğaziçi University, see: <http://bit.ly/bounethics>

The students of the Department of Economics are also informed about the consequences of academic dishonesty at the EC 103 Orientation to Economics course. For a brief refresher, see <http://bit.ly/ec103notes>

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Tentative Course Schedule

The material that will be covered is summarized below. In the course of progress this schedule may change. We will try to notify you in advance if and when such a change occurs.

Part I by Murat Koyuncu

Course Description

The first part of the course has two main objectives: (i) to further develop the modeling techniques that students acquire in the first year graduate macroeconomics courses, in terms of growth models with heterogeneous agents and structural change models and (ii) to familiarize the students with the recent developments in dynamic macroeconomics literature, specifically the structural change literature. The emphasis will be given to the second objective.

Course Reading List

A. Heterogeneous Agent Models – Complete Asset Markets

A1. Basic Models:

Caselli, F., and Ventura, J. (2000). "A representative consumer theory of distribution," *American Economic Review*, Vol. 90, pp. 909-926.

Chatterjee, S. (1994). "Transitional dynamics and the distribution of wealth in a neoclassical growth model", *Journal of Public Economics*, Volume 54, Issue 1, pp. 97-119.

García-Peñalosa, C. and Turnovsky, S. J. (2006). "Growth and Income Inequality: A Canonical Model," *Economic Theory*, Vol. 28, No. 1, pp. 25-49.

Ogaki, M. (2003). "Aggregation under Complete Markets", *Review of Economic Dynamics*, Volume 6, Issue 4, pp. 977-986.

A2. Some Applications

Alonso-Carrera J., J. Caballé, and X. Raurich, (2005). "Growth, habit formation and catching-up with the Joneses," *European Economic Review*, Vol. 49, pp. 1665-1691.

Alvarez-Cuadrado, F., G. Monteiro, and S.J. Turnovsky, (2004). "Habit formation, catching up with the Joneses, and non-scale growth," *Journal of Economic Growth*, 9, 47-80.

Boppart, T. and Ngai, L.R. (2017). "Rising inequality and trends in leisure", *mimeo*.

Carroll, C.D., J.R. Overland, and D.N. Weil, (1997). "Comparison utility in a growth model," *Journal of Economic Growth*, 2, 339-367.

De Nardi, M. and Fella, G. (2017). "Saving and Wealth Inequality," *mimeo*.

Dupor, B. and Liu, W.F. (2003). "Jealousy and equilibrium overconsumption," *American Economic Review*, Vol. 93, pp. 423-428.

García-Peñalosa, C. and Turnovsky, S. J. (2007). "Growth, Income Inequality, and Fiscal Policy: What Are the Relevant Trade-offs?," *Journal of Money, Credit and Banking*, Vol. 39, pp. 369-394.

García-Peñalosa, C. and S. J. Turnovsky, (2008). "Consumption externalities: A representative consumer model when agents are heterogeneous," *Economic Theory* 37, 439-467.

Liu, W. and Sarte, P.- D. G. (2004). "Progressive Taxation and Long-Run Growth," *American Economic Review*, 94(5), 1705-1716.

B. Structural Change

B1. Basic Models:

Acemoglu, D. and V. Guerrieri (2008). "Capital Deepening and Nonbalanced Economic Growth," *Journal of Political Economy*, 116, 467-498.

Caselli, F., and Coleman, W.J. (2001). "The U.S. structural transformation and regional convergence: a reinterpretation", *Journal of Political Economy* 109, 584–616.

Kongsamut, P., S. Rebelo and D. Xie (2001). "Beyond Balanced Growth," *Review of Economic Studies*, 68, 869-882.

Ngai L. R. and C. A. Pissarides (2007). "Structural Change in a Multisector Model of Growth," *American Economic Review*, 97, 429-443.

B2. Recent Literature:

Acemoğlu, D. and Autor, D. (2010). "Skills, Tasks and Technologies: Implications for Employment and Earnings," NBER Working Paper 16082.

Adão, R., Berajaz, M. and Pandalai-Nayar, N. (2019) "Skill-biased Technological Transitions," *mimeo*.

Alder, S., Boppart, T. and Müller, A. (2018). "A theory of structural change that can fit the data", *mimeo*.

Alonso-Carrera, J., Felice, G. and Raurich, X. (2018). "Inequality and structural change under non-linear Engels' curves" UBEconomics Working Papers E18/374.

Alonso-Carrera, J., Freire-Serén, M.J. and Raurich, X. (2017). "Anatomizing the mechanics of structural change," UBEconomics Working Papers E17/360.

Alonso-Carrera, J. and Raurich, X. (2018). "Labor mobility, structural change and economic growth", *Journal of Macroeconomics* 56, 292-310.

Alvarez-Cuadrado, F., Van Long, N. and Poschke, M. (2018). "Capital-Labor Substitution, Structural Change and Growth," *Journal of Economic Dynamics & Control*, 87, 206–231.

Bárány, Z.L. and Siegel, C. (2017). "Disentangling Occupation- and Sector-specific Technological Change", *mimeo*.

- Bárány, Z.L. and Siegel, C. (2018). "Job Polarization and Structural Change," *American Economic Journal: Macroeconomics*, 10(1): 57–89.
- Boppart, T. (2014). "Structural Change and the Kaldor Facts in a Growth Model with Relative Price Effects and Non-Gorman Preferences," *Econometrica*, Vol. 82, No. 6, 2167–2196.
- Bridgman, B., Duernecker, G. and Herrendorf, B. (2017) "Structural Transformation, Marketization, and Household Production around the World", *mimeo*.
- Buera, F.J., and J.P. Kaboski (2012a). "The Rise of the Service Economy," *American Economic Review*, 102(6), 2540- 2569.
- Buera, F.J., and J.P. Kaboski (2012b). "Scale and the Origins of Structural Change," *Journal of Economic Theory*, 147, 684-712.
- Buera, F.J., J.P. Kaboski and Rogerson, R. (2015). "Skill Biased Structural Change," NBER Working Paper 21165.
- Bustos, P., Monras, J., Vincenzi, J.M.C. and Ponticelli, J. (2019). "Structural Transformation, Industrial Specialization, and Endogenous Growth," *mimeo*.
- Cardi, O. and Restout, R. (2015) "Imperfect mobility of labor across sectors: a reappraisal of the Balassa–Samuelson effect", *Journal of International Economics*, 97, 249–265.
- Caron, J., Fally, T. and Markusen, J. (2017). "Per Capita Income and the Demand for Skills", *mimeo*.
- Comin, D., Lashkari, D. and Mestieri, M. (2015). "Structural Change with Long-run Income and Price Effects," *mimeo*.
- Chen, C. (2017). "Untitled Land, Occupational Choice, and Agricultural Productivity", *American Economic Journal: Macroeconomics*, 9(4): 91–121.
- Dennis, B. N. and İşcan T.B. (2009). "Engel versus Baumol: Accounting for structural change using two centuries of U.S. data," *Explorations in Economic History*, 46, 186–202.
- Eckert F. and Peters, M. (2018). "Spatial Structural Change", *mimeo*.
- Ferreira, A. M. (2019). "Skill-Biased Technological Change and Inequality in the U.S.," MPRA Paper No. 93914.
- Gangopadhyay, K. and Mondal, D. (2019) "Agricultural Productivity, Relative Sectoral Prices and TFPs: Theory and Evidence," *mimeo*.
- Herrendorf, B., C. Herrington, and Valentinyi, A. (2015). "Sectoral Technology and Structural Transformation," *American Economic Journal: Macroeconomics*, 7(4), 104-133.
- Herrendorf, B., R. Rogerson, and Valentinyi, Á. (2013). "Two Perspectives on Preferences and Structural Transformation," *American Economic Review*, 103(7), 2752-2789.
- Herrendorf, B., Rogerson, R. and Valentinyi, Á. (2018). "Structural Change in Investment and Consumption: A Unified Approach," NBER Working Paper 24568.
- Herrendorf, B. and Schoellman, T. (2018). "Wages, Human Capital, and Barriers to Structural Transformation," *American Economic Journal: Macroeconomics*, 10(2): 1–23.

- Hobijn, B., Schoell, T. and Vindas Q., A. (2018). “Structural Transformation by Cohort,” *mimeo*.
- Kuralbayeva, K. and Stefanski, R. (2013). “Windfalls, structural transformation and specialization,” *Journal of International Economics*, 90, 273–301.
- Lagakos, D. and Michael E. Waugh (2013). “Selection, Agriculture, and Cross-Country Productivity Differences,” *American Economic Review*, Vol. 103, No. 2, pp. 948-980.
- Matsuyama, K. (2017). “Engel's Law in the Global Economy: Demand-Induced Patterns of Structural Change, Innovation, and Trade,” CEPR Discussion Paper, 12387.
- Michelacci, C., and Pijoan-Mas, J. (2016). “Labor Supply with Job Assignment under Balanced Growth,” *Journal of Economic Theory*, 163, 110-140.
- Porzio, T. and Santangelo, G. (2017). “Structural Change and the Supply of Agricultural Workers”, *mimeo*.
- Swiecki, T. (2017). “Determinants of structural change,” *Review of Economic Dynamics*, 24, 95–131.
- Uy, T., Yi, K.-M. and Zhang, J. (2014). “Structural Change in an Open Economy,” *Journal of Monetary Economics*, 60(6), 667-682.
- Young, A. (2014). “Structural Transformation, the Mismeasurement of Productivity Growth, and the Cost Disease of Services”, *American Economic Review*, 104(11): 3635–3667.

Part II by Meltem Poyraz

Course Description

The second part of the course aims to familiarize students with the macroeconomic models that are built to account for differences across firms in various dimensions and to explain how decisions of heterogeneous firms affect the aggregate economy. To this end we will go over a variety of macroeconomic studies that are related to labor, reallocation, innovation, trade and finance literatures.

Course Outline

I. Introduction to Macroeconomics and Firm Heterogeneity

- Kehoe, P. J., V. Midrigan, and E. Pastorino (2018). “Evolution of Modern Business Cycle Models: Accounting for the Great Recession,” *Journal of Economic Perspectives* 32(3), 141-166.
- Krusell, P. and A. A. Smith, Jr. (2006). “Quantitative Macroeconomic Models with Heterogeneous Agents,” *Advances in Economics and Econometrics*, Ninth World Congress, Blundell, R., Newey, W., Persson, T. (Eds.), Cambridge University Press, Cambridge.
- Syverson, C. (2011). “What Determines Productivity?” *Journal of Economic Literature* 49(2), 326–365.
- Terry, S. J. (2017). “Alternative Methods for Solving Heterogeneous Firm Models,” *Journal of Money, Credit and Banking* 49(6), 1081-1111.

2. Macroeconomics of Labor Markets

Main papers

- Davis, S. J., R. J. Faberman, J. Haltiwanger (2013). "The Establishment-Level Behavior of Vacancies and Hiring," *Quarterly Journal of Economics* 128(2), 581-622.
- Elsby, M. W. L. and R. Michaels (2013). "Marginal Jobs, Heterogeneous Firms, and Unemployment Flows," *American Economic Journal: Macroeconomics* 5(1), 1-48.
- Fujita, S. and M. Nakajima (2016). "Worker Flows and Job Flows: A Quantitative Investigation," *Review of Economic Dynamics* 22, 1-20.
- Kaas, L., and P. Kircher (2015). "Efficient Firm Dynamics in a Frictional Labor Market," *American Economic Review* 105(10), 3030-60.
- Krusell, P., T. Mukoyama and A. Sahin (2010). "Labour-Market Matching with Precautionary Savings and Aggregate Fluctuations," *Review of Economic Studies* 77(4), 1477-1507.

Other papers

- Davis, S. J., R. J. Faberman, and J. Haltiwanger (2006). "The Flow Approach to Labor Markets: New Data Sources and Micro-Macro Links," *Journal of Economic Perspectives* 20(3), 3-26.
- Elsby, M. W. L., R. Michaels and D. Ratner (2015). "The Beveridge Curve: A Survey," *Journal of Economic Literature* 53(3), 571-630.
- Wright, R., P. Kircher, B. Julien, and V. Guerrieri (2017). "Directed Search: A Guided Tour," NBER Working Paper No. 23884.

3. Reallocation and Firm Dynamics

Main papers

- Hopenhayn, H. A. (1992). "Entry, Exit and Firm Dynamics in the Long Run Equilibrium," *Econometrica* 60(5), 1127-1150.
- Hopenhayn, H. A., and R. Rogerson (1993). "Job Turnover and Policy Evaluation: A General Equilibrium Analysis," *Journal of Political Economy* 101(5), 915-938.
- Hsieh, C. and P. J. Klenow (2009). "Misallocation and Manufacturing TFP in China and India," *Quarterly Journal of Economics* 124(4), 1403-48.
- Restuccia, D. and R. Rogerson (2008). "Policy Distortions and Aggregate Productivity with Heterogeneous Establishments," *Review of Economic Dynamics* 11(4), 707-720.

Other papers

- Lee, Y., and T. Mukoyama (2018). "Entry, Exit and Plant-level Dynamics over the Business Cycle," *Journal of Economic Dynamics and Control* 96, 1-25.
- Luttmer, Erzo G. J. (2007). "Selection, Growth, and Size Distribution of Firms," *Quarterly Journal of Economics* 122, 1103-1144.
- Mukoyama, T. and S. Osotimehin (2019). "Barriers to Reallocation and Economic Growth: The Effects of Firing Costs," *American Economic Journal: Macroeconomics*, Forthcoming.

4. Heterogeneous firms and trade

Main papers

- Bernard, A. B., J. B. Jensen, S. J. Redding and P. K. Schott (2012). "The Empirics of Firm Heterogeneity and International Trade," *Annual Review of Economics* 4, 283-313.
- Melitz, M. J. (2003). "The Impact of Trade of Intra-Industry Reallocations and Aggregate Industry Productivity," *Econometrica* 71(6), 1695-1725.
- Demidova, S. and A. Rodriguez-Clare (2013). "The simple analytics of the Melitz model in a small economy," *Journal of International Economics* 90(2), 266-272.

Other Papers

- DeLoecker, J. and F. Warzynski (2012). "Markups and Firm-Level Export Status," *American Economic Review* 102 (6), 2437-2471.
- Demidova, S. (2017). "Trade Policies, Firm Heterogeneity, and Variable Markups," *Journal of International Economics* 108, 260-273.
- Edmond, C., V. Midrigan, and D. Xu (2015). "Competition, Markups and the Gains from International Trade," *American Economic Review* 105(10), 3183-3221.
- Helpman, E., M. J. Melitz and S. R. Yeaple (2004). "Export versus FDI with Heterogeneous Firms," *American Economic Review* 94(1), 300-316.
- Helpman, E., M. J. Melitz and Y. Rubinstein (2008). "Estimating Trade Flows: Trading Partners and Trading Volumes," *Quarterly Journal of Economics* 123(2), 441-487.

5. Innovation and firm dynamics

Main papers

- Klette, T. J. and S. Kortum (2004). "Innovating Firms and Aggregate Innovation," *Journal of Political Economy* 112(5), 986-1018.

Other papers

- Lentz, R. and D. Mortensen (2005). "Productivity Growth and Worker Reallocation," *International Economic Review* 46(3), 731-749.
- Lentz, R. and D. Mortensen (2008). "An Empirical Model of Growth through Product Innovation," *Econometrica* 76(6), 1317-1373.
- Acemoglu, D., U. Akcigit, H. Alp, N. Bloom, W. R. Kerr (2018). "Innovation, reallocation and growth," *American Economic Review* 108(11) 3450-91.
- Akcigit, U., and W. R. Kerr (2018). "Growth through Heterogeneous Innovations," *Journal of Political Economy* 126 (4), 1374-1443.

6. The aggregate implications of firm level financial frictions

Main Papers

- Buera, F. J., J.P. Kaboski, and Y. Shin, (2011). "Finance and Development: A Tale of Two Sectors," *American Economic Review* 101(5), 1964-2002.
- Cabral, L.M.B, and J. Mata, (2003). "On the Evolution of the Firm Size Distribution: Facts and Theory," *American Economic Review* 93(4), 1075-1090.
- Caggese, A., and V. Cuñat, (2013). "Financing Constraints, Firm Dynamics, Export Decisions, and Aggregate Productivity," *Review of Economic Dynamics* 16(1) (Special Issue on Misallocation and Productivity), 177-193.

- Clementi, G., and H. Hopenhayn (2006). "A Theory of Financing Constraints and Firm Dynamics," *Quarterly Journal of Economics* 121(1), 229- 265.
- Jermann, U. J. and V. Quadrini (2007). "Stock market boom and the productivity gains of the 1990s," *Journal of Monetary Economics* 54(2), 413-432.

Other Papers

- Gilchrist, S., J. Sim and E. Zakrajsek (2013). "Misallocation and Financial Frictions: Some Direct Evidence from the Dispersion in Borrowing Costs," *Review of Economic Dynamics* 16(1), 159-176.
- Manova, K. (2013). "Credit Constraints, Heterogeneous Firms, and International Trade," *Review of Economic Studies* 80, 711-744.
- Oberfield, E. (2013). "Productivity and misallocation during a crisis: Evidence from the Chilean crisis of 1982," *Review of Economic Dynamics* 16(1), 100–119.

7. Finance, uncertainty, firm dynamics and the business cycle: theory and applications to the 2007- 2009 recession

Main Papers

- Arellano, C., Y. Bai, and P. Kehoe (2019). "Financial Markets and Fluctuations in Volatility", *Journal of Political Economy*, Forthcoming.
- Christiano, L. J., R. Motto, and M. Rostagno (2014). "Risk Shocks," *American Economic Review* 104(1), 27-65.
- Gilchrist, S., Sim, J., and Zakrajsek, E. (2014). "Uncertainty, Financial Frictions and Investment Dynamics," NBER Working Paper No. 20038.
- Jermann, U. J. and V. Quadrini (2012). "Macroeconomic Effects of Financial Shocks", *American Economic Review* 102(1), 238-271.
- Khan, A., and J. K. Thomas, (2013). "Credit Shocks and Aggregate Fluctuations in an Economy with Production Heterogeneity," *Journal of Political Economy* 121(6), 1055-1107.

Other Papers

- Bloom, N. (2009). "The Impact of Uncertainty Shocks," *Econometrica* 77(3), 623-686.
- Bloom, N., M. Floetotto, I. Saporta-Eksten, N. Jaimovich, and S. Terry (2018). "Really Uncertain Business Cycles," *Econometrica* 86(3), 1031-1065.
- Gilchrist, S. and E. Zakrajsek (2012). "Credit Spreads and Business Cycle Fluctuations," *American Economic Review* 102(4), 1692-1720.
- Gilchrist, S., R. Schoenle, J. Sim, and E. Zakrajšek (2017). "Inflation Dynamics during the Financial Crisis," *American Economic Review* 107(3), 785-823.
- Guerrieri, V. and G. Lorenzoni (2017). "Credit Crises, Precautionary Savings, and the Liquidity Trap," *Quarterly Journal of Economics* 132(3), 1427–1467.
- Monacelli, T., V. Quadrini, and A. Trigari (2011). "Financial Markets and Unemployment," NBER Working Paper No. 17389.
- Schaal, E. (2017). "Uncertainty and Unemployment," *Econometrica*, 85(6), 1675-1721.